

Technical Data · Animal Feed Applications

# Oceana Aragonite Feed Grade

Oolitic aragonite — the world's only recent-origin, renewable and sustainable performance mineral. Exceptionally pure calcium carbonate (aragonite calcium carbonate).

We harvest oolitic aragonite that is used as the feed material in screening and drying operations to produce grades of feed-grade aragonite. An all-natural product that can benefit the animal feed industry because of its characteristics and the benefits they lend to the industry. The higher solubility of aragonite versus calcite, in combination with an extremely large surface area associated with oolites and an extremely pure material (>39% calcium), provides an efficient delivery system for calcium to livestock.

## Solubility and crystalline structure

The solubility of aragonite is determined by its unique crystalline structure. Acid neutralization curves for oolitic aragonite were developed by the Keck Geophysical Laboratory at Stevens Institute of Technology and compared with an equilibrium curve developed using Visual Minteq Chemical Modeling Software (KTH, 2010) for calcite/limestone. Visual Minteq uses the process and chemical databases provided by the USEPA model MINTEQA2 (USEPA, 2010).

Results clearly indicate that use of aragonite supplies greater concentrations of calcium over a pH range of 5–7. This implies that at a pH of 5.5, it will require far less aragonite than calcite to deliver the same dose of calcium to animals. The aragonite measurements were completed on unaltered oolites with a particle size in the range of 150–350 microns and compared to an idealized, infinitely small particle size, 100% pure calcite. The surface area of the oolitic aragonite has been measured in excess of 1.7 m<sup>2</sup>/g (approximately 124,667 in<sup>2</sup>/in<sup>3</sup>).

## Aragonite composition — large surface area and porosity

| Property          | Oolitic Aragonite      | Calcite / Limestone    |
|-------------------|------------------------|------------------------|
| Specific Gravity  | 2.93+                  | 2.7                    |
| Mohs Hardness     | 3.5–4                  | 3                      |
| Crystal Structure | Orthorhombic           | Hexagonal              |
| Surface Area      | 1.82 m <sup>2</sup> /g | 0.55 m <sup>2</sup> /g |
| Zeta Potential    | –6.65 mV to 33.85 mV   | –1.01 mV to +11.55 mV  |
| Crystallinity     | High                   | Low                    |
| Micro Porosity    | Very High              | Low                    |

All information provided and recommendations made herein are intended to assist customers in determining whether our products are suitable for their applications. We request that customers inspect and test our products before use in order to make their own final decision regarding suitability. We do not guarantee results, freedom from patent infringement, or suitability of

resultant products for any suggested application with respect to the use of any formula or material described herein.